

Assignment, WASP Software Engineering Course Module 2026

This assignment requires you to write an essay on Software Engineering for AI/AS (“AI Engineering”), how it relates to your research, and how you can apply SE principles and tools in your project/sub-topic. Diagrams or illustrations are allowed if necessary (they don’t count toward page length).

Length and Format:

- 4 – 6 A4 pages of text (excluding references)
- Times New Roman (or equivalent), 11 pt. Be sensible in formatting / layout choices.
- Submit both Markdown (.md) or LaTeX (.tex) and PDF via Pull Request to the course repo: assignments/2026/surname_firstname/...
- **Deadline: 14:00 September 1, 2026**

The essay should be structured as follows:

1. Introduction

Briefly introduce your research area and topic so the rest of your essay stands on its own. (Maximum 500 words)

2. Lecture principles

- Pick two principles/ideas/concepts/techniques from Robert’s lectures or the course discussion sessions.
- Discuss how each relates to your research and topic.

3. Guest-lecture principles

- Pick two principles/ideas/concepts/techniques from the guest lectures.
- Discuss how each relates to your research and topic.

4. Data scientists, software engineers, and AI engineers

Read chapters “1. Introduction” and “2. From Models to Systems” of the CMU “Machine Learning in Production” book: - <https://mlip-cmu.github.io/book/01-introduction.html> - <https://mlip-cmu.github.io/book/02-from-models-to-systems.html>

Then answer: - What differences do the chapters describe between data scientists and software engineers? Do you agree with this distinction? Why or why not? - The chapters argue that production ML requires a system-wide view, not only a model-centric view. How does this apply to your own research area? - Do you think data scientists and software engineers will remain distinct roles, or will both need to learn more of each other’s skills? Explain. - Where does “AI

Engineering” fit in this picture: is it mostly software engineering for AI systems, data science made more systematic, MLOps/model engineering, or something partly distinct? - How do foundation models, AI coding assistants, RAG systems, prompt pipelines, and agentic workflows change the boundaries between these roles?

5. Paper analysis

Find and read two full/long papers:

- One full/long paper from any CAIN conference (links at <https://conf.researchr.org/series/cain>).
- One other recent full/long paper, preferably from 2024-2026 (can be but doesn't have to be from CAIN conference), related to one of the 2026 course themes: LLMs for software engineering, AI-assisted development, LLM/RAG/prompt evaluation, testing of ML/AI systems, agentic software engineering, AI system security/safety, technical debt in AI systems, or software engineering for foundation-model-based systems.

For each paper cover:

1. Core ideas and their SE importance
 - Describe the core idea(s) of the paper and why it/they are important to the engineering of AI systems
2. Relation to your research
 - How the paper relates to your own research
3. Integration into a larger AI-intensive project
 - Describe a (real or fictional) larger AI-intensive software project, where AI/ML is a core aspect of the system's functionality
 - Describe how the paper's ideas could be used in this project
 - How your own WASP research could fit into this project
4. Adaptation of your research
 - How you might tweak your research project to (long-term) better support the paper's AI-engineering idea and the challenges it is (trying to) address

Your answers to this section should be approximately 2 A4 pages in length, about 1 page per paper.

6. Evidence, originality, and GenAI-use transparency

Briefly explain what sources you used and how you used them:

- List the main papers and sources you relied on.
- For each main source, state what kind of source it is, for example peer-reviewed paper, preprint, book chapter, standard/framework, product page, or blog post.
- Comment briefly on which sources you trust most and why.

- If you used GenAI tools for brainstorming, search help, language checking, or outlining, briefly disclose how.
- Do not submit generated text as your own analysis; you are responsible for all claims, citations, and wording.

Submission

Submit your report in both raw/text format (preferably Markdown (.md) or LaTeX (.tex), contact Robert on email if you have reasons to not use any of them) and as a PDF by creating a Pull Request to the course git repo on GitHub: https://github.com/robertfeldt/wasp_SE_course

Create your own **directory** named “surname_firstname” (example for me would be “feldt_robert”) under the

https://github.com/robertfeldt/wasp_SE_course/tree/main/assignments/2026 directory and then put your assignment files (see above) in your directory and submit as a Pull Request.

Deadline for submission of report files: At **14:00 on September 1st 2026**. If you are graduating before October 2026 and need to be sure to have your course credits email Robert and he can try to prioritize the grading of your assignment; no guarantees though (due to vacation time and many students taking the course).

Assessment

Assessment will be light, and reports will be judged either “pass” or “fail”. A pass will occur when your report is deemed to show original work and thought, is of the required length, engages seriously with the course material, and relates the ideas to your own research. Do not copy/paste from any source, even if you are referring to a paper/article. Do not submit GenAI-generated text as your own analysis. If originality is unclear, we may ask for clarification or an additional oral discussion.

Other information

This is intended as a light piece of work – do not overthink it, but study the course material, find relevant papers, then think and relate to your own project and experience, then provide your thoughts and perspectives. Most important is that you show you have gained some insight and have related the course material to your own project and research.

If you have any questions email robert.feldt@chalmers.se